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BMATE101 – MATHEMATICS-I (EEE)

MODULE-4 : INTEGRAL CALCULUS

DOUBLE INTEGRALS & CHANGE OF ORDER

1. Double Integrals

Definition

A **double integral** is used to integrate a function of two variables over a region R in the xy -plane.

$$\iint_R f(x, y) \, dx \, dy$$

It represents:

- Area (if $f(x, y) = 1$)
- Volume (if $f(x, y) \neq 1$)

2. Double Integral as an Iterated Integral

If region R is bounded by:

$$a \leq x \leq b, g_1(x) \leq y \leq g_2(x)$$

Then:

$$\iint_R f(x, y) \, dx \, dy = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$

Example 1

Evaluate:

$$\int_0^1 \int_0^2 (x + y) \, dy \, dx$$

Solution

Inner integral:

$$\int_0^2 (x + y) dy = 2x + 2$$

Outer integral:

$$\int_0^1 (2x + 2) dx = 3$$

3. Change of Order of Integration

Sometimes changing the order:

$$dy dx \leftrightarrow dx dy$$

makes integration easier.

Procedure

1. Sketch the region of integration
2. Rewrite limits in the new order
3. Integrate

Example 2

Change the order of integration:

$$\int_0^1 \int_0^x f(x, y) dy dx$$

Region:

$$0 \leq y \leq x \leq 1$$

New order:

$$\boxed{\int_0^1 \int_y^1 f(x, y) dx dy}$$

4. Double Integrals in Polar Coordinates

When the region is circular, conversion to **polar coordinates** simplifies integration.

Transformation

$$x = r \cos \theta, y = r \sin \theta$$

Jacobian:

$$\boxed{dx dy = r dr d\theta}$$

Example 3

Evaluate:

$$\iint_R (x^2 + y^2) dx dy$$

over the circle $x^2 + y^2 = a^2$.

Convert:

$$\begin{aligned} x^2 + y^2 &= r^2 \\ \int_0^{2\pi} \int_0^a r^3 dr d\theta \\ &= \frac{\pi a^4}{2} \end{aligned}$$

5. Applications of Double Integrals

5.1 Area

If $f(x, y) = 1$:

$$\text{Area} = \iint_R dx dy$$

5.2 Volume

If surface $z = f(x, y)$:

$$\boxed{\text{Volume} = \iint_R f(x, y) dx dy}$$

Example 4 (Area)

Area of region bounded by:

$$\begin{aligned} x &= 0, y &= 0, x &+ y = 1 \\ \text{Area} &= \int_0^1 \int_0^{1-x} dy dx = \frac{1}{2} \end{aligned}$$

TRIPLE INTEGRALS, BETA & GAMMA FUNCTIONS

6. Triple Integrals

Definition

A **triple integral** is used to integrate a function of three variables over a region V in space.

$$\iiint_V f(x, y, z) \, dx \, dy \, dz$$

It is mainly used to find:

- **Volume**
- **Mass**
- **Centroid** (self-study)

7. Triple Integral as Iterated Integral

If region V is bounded by:

$$a \leq x \leq b, g_1(x) \leq y \leq g_2(x), h_1(x, y) \leq z \leq h_2(x, y)$$

Then:

$$\iiint_V f(x, y, z) \, dx \, dy \, dz = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{h_1(x, y)}^{h_2(x, y)} f(x, y, z) \, dz \, dy \, dx$$

Example 1

Evaluate:

$$\iiint_V 1 \, dx \, dy \, dz$$

where:

$$\begin{aligned} 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1 \\ = \int_0^1 \int_0^1 \int_0^1 1 \, dz \, dy \, dx = 1 \end{aligned}$$

Hence, **volume of the cube = 1.**

8. Change of Variables in Triple Integrals (Brief)

When symmetry exists, change of variables simplifies integration.

Example (cylindrical):

$$x = r \cos \theta, y = r \sin \theta$$

Jacobian:

$$dx dy dz = r dr d\theta dz$$

(This is sufficient as per syllabus.)

9. Beta Function

Definition

The **Beta function** is defined as:

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

where:

$$m > 0, n$$

$$> 0$$

Properties of Beta Function

1. Symmetry:

$$\beta(m, n) = \beta(n, m)$$

2. Relation with Gamma function (important):

$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

10. Gamma Function

Definition

The **Gamma function** is defined as:

$$\Gamma(n) = \int_0^\infty x^{n-1} e^{-x} dx \quad (n > 0)$$

Important Results

1. Recurrence relation:

$$\Gamma(n+1) = n\Gamma(n)$$

2. For positive integers:

$$\Gamma(n) = (n-1)!$$

3. Special value:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

11. Relation Between Beta and Gamma Functions

$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

This relation is **frequently used** in problems.

12. Problems on Beta & Gamma Functions

Example 2

Evaluate:

$$\beta(2,3)$$

Using relation:

$$\begin{aligned}\beta(2,3) &= \frac{\Gamma(2)\Gamma(3)}{\Gamma(5)} \\ &= \frac{1! \times 2!}{4!} = \frac{2}{24} = \frac{1}{12}\end{aligned}$$

Example 3

Evaluate:

$$\begin{aligned}\Gamma\left(\frac{5}{2}\right) \\ \Gamma\left(\frac{5}{2}\right) &= \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{3}{4}\sqrt{\pi}\end{aligned}$$

13. Applications of Module–4

- Volume computation
- Mass distribution
- Probability distributions
- Electrical and electromagnetic field analysis
- Wave propagation